

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section.

FR-DLT-001: Load Customer and Order Data from CSV to Delta Tables

Title: Load Customer and Order Data from CSV to Delta Tables

Description: The system must read customer and order data from CSV files and load it into Delta tables, performing data cleansing and transformation as required.

Preconditions:

CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables `customer` and `order` exist or are created

Main Flow / Functional Steps:

1. Read customer data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into Delta table `customer`.
2. Read order data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into Delta table `order`.
3. Add a new column `TotalAmount` to the `order` table, calculated as the product of `PricePerUnit` and `Qty`.
4. Remove null and duplicate records from both `customer` and `order` tables.

FR-DLT-002: Create and Load Order Summary Table

Title: Create and Load Order Summary Table

Description: The system must create an `ordersummary` table if it does not exist and load it with joined data from `customer` and `order` tables using SCD Type 2 logic.

Preconditions:

- `customer` and `order` tables are loaded with data
- `ordersummary` table does not exist or is created

Main Flow / Functional Steps:

1. Create `ordersummary` table if it does not exist in catalog `gen\_ai\_poc\_databrickscoe` and schema `sdlc\_wizard`.

2. Join `customer` and `order` tables on `CustId` column.
3. Load joined data into `ordersummary` table using SCD Type 2 logic, maintaining history.

FR-DLT-003: Update Order Summary Table on Customer Changes

Title: Update Order Summary Table on Customer Changes

Description: The system must update the `ordersummary` table whenever there are changes to the `customer` table, using SCD Type 2 logic.

Preconditions:

- `customer` table is updated
- `ordersummary` table exists and is loaded with data

Main Flow / Functional Steps:

1. Detect changes to the `customer` table.
2. Update `ordersummary` table by making old records inactive and new records active, updating `StartDate` and `EndDate` accordingly.

FR-DLT-004: Create and Load Customer Aggregate Spend Table

Title: Create and Load Customer Aggregate Spend Table

Description: The system must create a `customeraggregatespend` table if it does not exist and load it with aggregated data from the `ordersummary` table.

Preconditions:

- `ordersummary` table exists and is loaded with data
- `customeraggregatespend` table does not exist or is created

Main Flow / Functional Steps:

1. Create `customeraggregatespend` table if it does not exist in catalog `gen\_ai\_poc\_databrickscoe` and schema `sdic\_wizard`.
2. Aggregate `TotalAmount` column from `ordersummary` table by `Name` and `Date` columns.
3. Load aggregated data into `customeraggregatespend` table.

FR-DLT-005: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements are implemented

Main Flow / Functional Steps:

1. Implement all previous requirements using Delta Live Tables.

Let me know if this meets your expectations or if you need any further clarification.